# Learning to Manipulate a New Object from Action-Free Videos

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ECSE 552 — Final report (draft)

## Abstract

Visuomotor policies imitate images paired with motor commands, and every new object normally requires new paired demonstrations. We study whether videos without actions are enough to extend an image-only policy from one object to another. In simulation (OCBench, UR5e arm), labeled demonstrations exist only for a red source cube; each target object has only videos. An inverse dynamics model (IDM), trained on the source and adapted to the target with domain-adversarial training (DANN), labels the target videos, and an image-only ACT policy trains on both. The reference policies are measured with three seeds each. The source-only policy reaches $90.7 \pm 12.1\%$ test success on the source cube, transfers to red blocks of another shape (82% and 59%) and never grasps a blue object (0.3% and 0%). Training the same policy with colour jitter, without any target data, closes that gap: 98.7% on the blue cube, above the oracle trained with the target's true actions (85.7%). Appearance shifts therefore do not need target videos. We added a target that changes the behaviour instead: a block that only fits the gripper after a $90^\circ$ turn, where both source-only policies fail (0% and 0.3%) and the oracle reaches 89.0%. [TODO: Results of the IDM methods against the baselines, R4 and R5, and the conclusion.]

## 1 Introduction

**Application (C1).** Warehouse and home robots meet new items constantly. A robot arm that already manipulates one object should learn to handle a new one without anyone operating the robot again. We study this for a tabletop pick-and-place task in the OCBench simulator [1], built on MuJoCo [2]: a UR5e arm with a Robotiq gripper picks up a block and places it inside a goal square, seeing the scene only through three RGB cameras (Figure 1).

**Why a solution is needed (C3).** Visuomotor policies such as ACT [3] and Diffusion Policy [4] imitate images paired with motor commands. Acquiring such pairs is costly because it requires operating the robot: 50 people took a year to collect the 76k teleoperated demonstrations of DROID [5]. With limited coverage, a policy that meets an unfamiliar situation may output an erroneous action, and errors compound during execution, moving the robot further from its training data [3]. Videos are far cheaper than paired data: they can be recorded without logging commands, and world models can now generate them [6]. The open question is how to turn videos into supervision for a policy that must output motor commands.

**Related work.** The standard recipe trains an *inverse dynamics model* (IDM) on transitions with known actions, uses it to label observation-only demonstrations, and clones the result: behavioral cloning from observation [7], scaled to internet video by VPT [8] and to video generated by world models by DreamGen [6]. Morin et al. [9] analyse why this is sample efficient with few labels and name unlabeled data from a different environment as future work; this is our setting. Kim et al. [10] label human eye-in-hand videos with an IDM trained on robot play. A second line avoids explicit

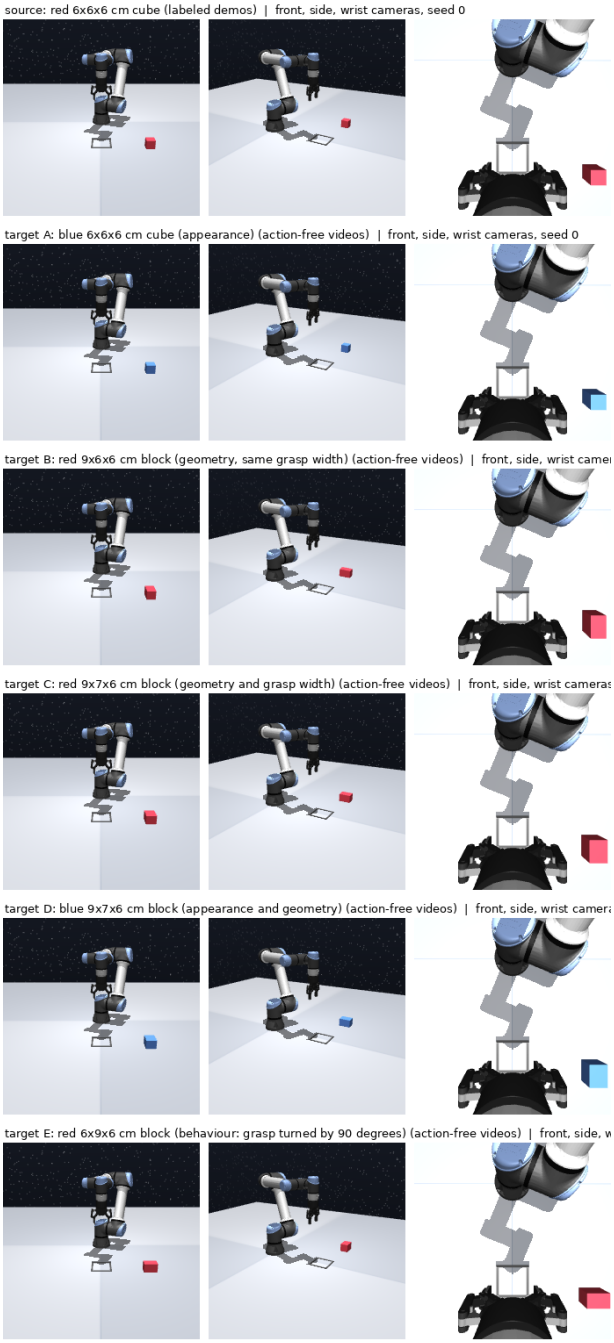



Figure 1: Initial state of every object from the same seed, seen by the front, side and wrist cameras. The gray square is the goal.

pseudo-actions by learning latent actions or motion representations from videos [11–13]; these methods need larger models and data than this project allows. When the videos show a new object, the IDM is applied outside its training distribution. Unsupervised domain adaptation addresses this shift by aligning source and target features, adversarially in DANN [14] and its successors [15, 16]. Alignment has been applied to the policy's encoder for visuomotor transfer [17, 18], but Cheng et al. [17] require action-labeled target demonstrations. Two warnings apply to our design: invariant features can hurt when the label distribution differs between domains [19], and adversarial alignment was designed for classification and is weaker for regression [20]. We found no work that adapts an

Table 1: Requirements (C4) and their status in this draft. R4 and R5 are filled for the augmentation baseline (e); the methods that use target videos are pending.

| ID | Requirement | Definition | Target | Status |
|---|---|---|---|---|
| R1 | Supervision | Target data are RGB frames only | Zero target labels | By design; [TODO: confirm for the IDM pipeline] |
| R2 | Label budget | Labeled source demonstrations, identical for every method | 100; 300 if the source-only policy is below 80% in source validation | Decided: 300 (100 demonstrations reach 37% in validation) |
| R3 | Inputs | No simulator state in any learned model | Images only | Met by the policy (robot state zeroed); [TODO: IDM] |
| R4 | Target | Success gain over the source-only policy on each target, as a fraction of the oracle's gain | $\geq 50\%$ of the gap | (e): Table 11; [TODO: (b), (c) per target] |
| R5 | Retention | Drop in source success after adding target data | $\leq 10$ points | (e): no drop (95.0% against 90.7%); [TODO: (b), (c) per target] |
| R6 | Real time | Policy inference per call, within one 10 Hz control step (IDM offline) | $< 100$ ms | Met: 10 to 18 ms (95th percentile) in the 60 test runs |

IDM adversarially to pseudo-label action-free videos of a new object for the same robot; we frame this as a controlled study and not as a new method.

**Problem (C2, high level).** Labeled demonstrations (images and actions) exist only for a *source* object. For each *target* object there are only videos. We want one image-only policy per target that succeeds on the target without losing the source, and we ask whether adapting the IDM to the target before labeling its videos helps.

**Scope.** The source is a $6 \times 6 \times 6$ cm red cube. Targets are (A) a blue cube of the same size (appearance), (B) a $9 \times 6 \times 6$ cm red block (geometry, same grasp width), (C) a $9 \times 7 \times 6$ cm red block (geometry and grasp width), (D) a blue version of C and (E) B's block lying across the gripper, which must be grasped with a $90^\circ$ turn (behaviour). E was added after the results of the augmentation baseline (Section 3.5). Robot, cameras, lighting and task stay fixed. Human videos, large pretrained models and real robots are out of scope. This draft has the source-only baseline and the augmentation baseline on every object, and the oracle for A, B, C and D. [TODO: Which targets the IDM methods cover.]

### 1.1 Constraints

Table 1 lists the requirements the design must satisfy (C4). R1–R3 define the supervision contract and hold by construction; R4 and R5 are the outcome targets; R6 is the real-time condition. R6 was restated when the control rate changed (Section 2.3): the policy is queried at every 10 Hz control step, so one call must finish within the 100 ms control period.

## 2 Methods

### 2.1 Problem formulation

An episode is a sequence of observations $o_t$ (three RGB images) and actions $a_t \in \mathbb{R}^7$ (six arm joints and the gripper). The source dataset contains labeled transitions, $\mathcal{D}_S = \{(o_t, a_t, o_{t+1})\}$, from $N_S$ demonstrations of the source object. Each target $T \in \{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}\}$ has only videos, $\mathcal{D}_T = \{o_t\}$. Source and target share the robot, the cameras, the task and the expert's strategy; they differ in the object's appearance or geometry, so the observation distribution shifts and, for geometry changes, the action distribution may shift as well. For target E it must: the grasp is turned.

Table 2: Objects and collections. Each target collection is 200 action-free videos; its stored actions serve only the oracle and evaluation. Demonstrations average 70 to 76 steps (about 7 s). The turned-grasp source collection trains only the IDM (Section 3.5).

| Object | Shape (cm), colour | Changes | Episodes | Seeds from | Expert success |
|---|---|---|---|---|---|
| Source | $6 \times 6 \times 6$, red | (labeled) | 300 | 0 | 99.7% |
| Source, held out | $6 \times 6 \times 6$, red | (IDM test only) | 50 | 50,000 | 100% |
| A | $6 \times 6 \times 6$, blue | Appearance | 200 | 100,000 | 100% |
| B | $9 \times 6 \times 6$, red | Geometry | 200 | 200,000 | 99.0% |
| C | $9 \times 7 \times 6$, red | Geometry, grasp width | 200 | 300,000 | 100% |
| D | $9 \times 7 \times 6$, blue | All three | 200 | 400,000 | 100% |
| E | $6 \times 9 \times 6$, red | Behaviour | 200 | 500,000 | 93.9% |
| Source, turned | $6 \times 6 \times 6$, red | (IDM only) | 100 | 600,000 | 100% |

We seek, for each target, an image-only policy $\pi_T(a_{t:t+H} \mid o_t)$ that predicts a chunk of $H$ future actions. Let $s(\pi, X)$ be the closed-loop success rate of policy $\pi$ on object $X$. Three reference policies anchor the requirements: $\pi_a$ trained on $\mathcal{D}_S$ only, $\pi_d$ trained on $\mathcal{D}_S$ plus the target with its true actions (oracle), and the candidate $\pi_T$ trained on $\mathcal{D}_S$ plus pseudo-labeled $\mathcal{D}_T$. R4 and R5 are

$$\text{R4:} \quad G_T = \frac{s(\pi_T, T) - s(\pi_a, T)}{s(\pi_d, T) - s(\pi_a, T)} \geq 0.5, \tag{1}$$

$$\text{R5:} \quad s(\pi_a, S) - s(\pi_T, S) \leq 0.10. \tag{2}$$

R4 is inconclusive for a target if the oracle does not exceed the source-only policy, $s(\pi_d, T) \leq s(\pi_a, T)$. True target actions are used only to train the oracle and to evaluate pseudo-label error; they never select a model (R1).

### 2.2 Datasets and resources

All data are generated by us in simulation; no licensing or ethics constraints apply.

**Scene and task.** We use OCBench's visual single-block task (`visual-block-cpu-lite-single-task1-v0`) through our fork of the simulator, which adds object colours and sizes, a visible goal and reproducible initial states. A dark-gray 12 cm square marks the fixed goal. An episode succeeds when the whole object is inside the square, resting on the table, released and still for 1 s; the time limit is 200 control steps (20 s). The object starts uniformly in a $20 \times 20$ cm area beside the goal with its yaw fixed at 0, and the arm always starts from the same pose. The 12 cm side was chosen from the expert's own precision: 97% of its resting placements fit fully inside a 12 cm square, against 60% for 8 cm (200 episodes).

**Observations and actions.** Each control step (10 Hz) stores one $224 \times 224$ RGB image from each of the front, side and wrist cameras. OCBench actions are seven values in $[-1, 1]$: six arm-joint increments and one gripper-opening increment, each added to the current measured position. Simulator state (joint positions and velocities) is stored for replay, evaluation and label conversion, but is never an input of a learned model (R3).

**Objects.** Only the manipulated object changes between domains (Table 2); robot, cameras, goal, initial states and expert settings are shared. Blocks lie with their 9 cm side along $x$ (along $y$, the closing direction of the gripper, for E), and success requires the whole block inside the 12 cm square. The gripper opens 8.5 cm, so the expert grasps every block across its shorter side, which leaves 1.25 cm of clearance per side for a 6 cm object and 0.75 cm for a 7 cm one. Targets C and D are grasped 2 cm above the centre: at centre height the gripper linkage hits a 7 cm block, and 47% of episodes needed a second grasp attempt, against 12% when grasping higher (the same rate as the source). For E the shorter side lies along $x$, so the expert turns the grasp by $90^\circ$ (Figure 2); its success is lower (93.9%) because the placed block ends $0.8 \pm 0.3$ cm off-centre along $y$, where the 9 cm side has 1.5 cm of room. Failed attempts are not kept.

**Collections and seeds.** Demonstrations come from OCBench's scripted expert. Every episode has its own seed, which fixes the initial state and the expert's random choices; failed attempts are

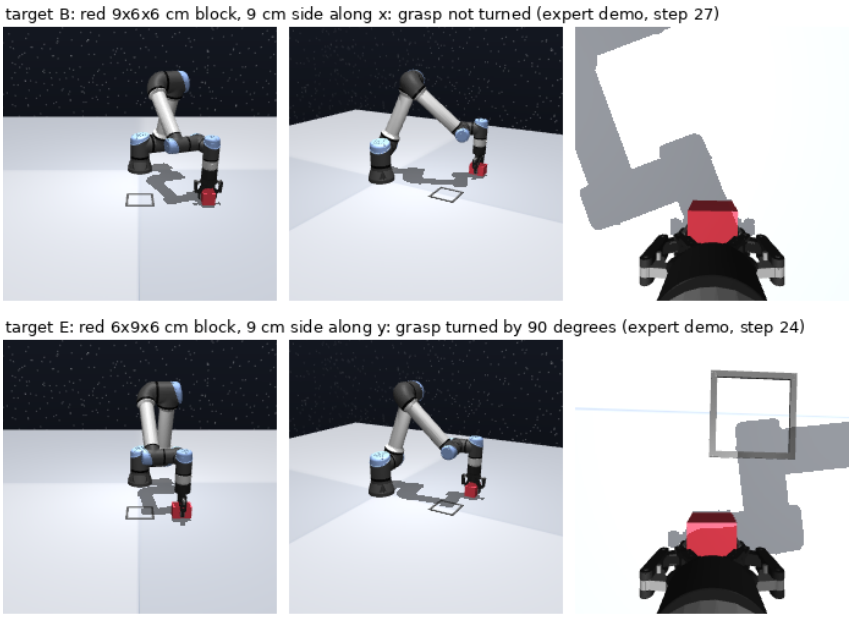



Figure 2: Expert grasp of target B (top) and of target E (bottom): the same block, lying along or across the gripper. Only E needs the grasp turned by 90°.

discarded and recorded. Seed ranges do not overlap: each collection has its own (Table 2), validation rollouts use seeds from 900,000 and test rollouts from 1,000,000. The same seed gives the same initial object and arm state for every object, so all methods are evaluated from identical initial states, and no evaluation start is a training start. We verified that stored episodes replay exactly from their actions.

**Deterministic against stochastic demonstrations.** Stock OCBench demonstrations are deliberately varied: the cube has a random yaw, and the expert moves at a random speed and takes deliberate detours. We collected both this *stochastic* variant and a *deterministic* one (yaw fixed at 0, no detours, fixed speed), with the same seeds. The same unmodified ACT, trained on 300 source demonstrations with one seed, reached 99% test success on the deterministic data and 53% on the stochastic data (Section 3.9). Evaluating the stochastic policy from fixed-yaw starts gave 71% in validation, so about 18 points of the gap come from the harder initial states and about 27 from the variability of the demonstrations. We therefore fixed the deterministic data as the project data for every method and object, before running any method that uses target videos: the research question concerns the transfer between objects, and a source-only policy that fails half the time on its own object would leave no room to measure retention (R5) or a meaningful oracle gap (R4). The stochastic collections are kept as a reference.

**No grasp turns.** The first deterministic collections kept one more source of variation: the stock expert turns its grasp by 90° in 30% of cube episodes. It can never do so for a block, because 9 cm does not fit the gripper, so the source and target B would have differed in more than geometry. We removed the turn for every object and recorded all six collections again with the same seeds. Target E, added later, is the one object that is always grasped turned. The rest of the expert's grasp randomness stays (a pick hold, about 1 cm of contact jitter, second attempts). This change was made after the R2 result below, which is a deviation from fixing the protocol before any result; the label budget was not revisited.

**Resources.** Episodes are public on the Hugging Face Hub (`https://huggingface.co/datasets/Autobrik/ecse552-ocbench-block`); training downloads a pinned revision, and checkpoints are published in companion model repositories. A LeRobot copy [21] serves the online visualizer. Policies train on Modal with one NVIDIA A10G GPU, about 2 to 3.5 h per ACT run (5 h with augmentation, which runs in ACT's single data-loading process), with

curves logged to Weights & Biases. Rendering and rollouts run on a local GPU. The code is at `https://github.com/BrikHMP18/ecse-552-project`.

### 2.3 Designed solution and implementation

The pipeline has three stages: (1) train an IDM on labeled source transitions, adapting it to the target with DANN; (2) label the target videos with the IDM; (3) train an image-only ACT policy on source demonstrations plus pseudo-labeled target videos. One policy is trained per target. IDM and policy share no weights.

**Inverse dynamics model.** The IDM $f_\theta$ predicts the action between two consecutive observations, $\hat{a}_t = f_\theta(o_t, o_{t+1})$. A convolutional encoder shared across cameras embeds both frames and their difference, which exposes motion and suppresses static appearance; an MLP regresses the action with a mean-squared-error loss on source transitions,

$$\mathcal{L}_{\text{act}}(\theta) = \mathbb{E}_{(o_t, a_t, o_{t+1}) \sim \mathcal{D}_S} \left\| f_\theta(o_t, o_{t+1}) - a_t \right\|_2^2. \tag{3}$$

The IDM is easier to learn from few labels than the policy because it does not need to model the expert's intent, only the effect of an action that already happened [9]. Its split takes blocks of ten source episodes, eight for training and two for validation. [TODO: IDM architecture (encoder, input resolution, output: absolute joint targets), optimizer, training length and source validation error.]

**Domain-adversarial adaptation.** Following DANN [14], a domain discriminator $d_\phi$ receives the IDM's features $h_\theta$ for source and target frame pairs and is trained with binary cross-entropy to tell them apart. A gradient reversal layer makes the encoder maximize the same loss, scaled by $\lambda$:

$$\min_\theta \max_\phi \;\; \mathcal{L}_{\text{act}}(\theta) - \lambda\, \mathcal{L}_{\text{dom}}(\theta, \phi), \qquad \mathcal{L}_{\text{dom}} = \text{BCE}\big(d_\phi(h_\theta(o_t, o_{t+1})),\, y_{\text{dom}}\big). \tag{4}$$

The encoder is thus pushed towards features that predict source actions but do not reveal the domain. $\lambda$ ramps up on a schedule fixed in advance, and $\lambda = 0$ gives the unadapted IDM of method (b). Only target *frames* enter this stage (R1). IDM checkpoints are selected by source validation error, never with target actions. [TODO: Discriminator architecture, $\lambda$ schedule and final value, batch composition.]

**Policy.** The policy is the original ACT [3], run from the authors' code without modification through a thin adapter. ACT is a conditional VAE: a ResNet-18 backbone per camera feeds a transformer encoder (4 layers) and decoder (7 layers, 8 heads, hidden size 512, feed-forward size 3,200) that outputs a chunk of $H = 20$ actions (2 s); a separate encoder maps the action chunk to a 32-dimensional latent $z$ during training, and $z = 0$ at test time. The loss is

$$\mathcal{L}_{\text{ACT}} = \left\| \hat{a}_{t:t+H} - a_{t:t+H} \right\|_1 + \beta\, D_{\text{KL}}\big(q(z \mid a_{t:t+H}, o_t) \,\|\, \mathcal{N}(0, I)\big), \qquad \beta = 10. \tag{5}$$

We keep the hyperparameters of the ACT repository (batch size 8, learning rate $10^{-5}$, weight decay $10^{-4}$, 2,000 epochs). The only change is that the robot-state input is set to zero, so the policy sees images only (R3).

**Actions and execution.** Policies are trained on absolute joint targets, computed from the stored increments and joint positions: $q_t^* = q_t + a_t \odot c$ for the arm, with $c$ the simulator's per-joint scale, and the clipped opening for the gripper. A low-level controller converts each target back into the increment the simulator expects. At test time the policy is queried at every step and the overlapping chunks are combined by ACT's temporal ensembling: the executed target is $\sum_i w_i \hat{q}_t^{(i)} / \sum_i w_i$ with $w_i = \exp(-k\, i)$, $k = 0.01$, where $i = 0$ is the oldest prediction for step $t$.

**Training on several collections.** For the methods that use a target, a run's episodes are the source demonstrations followed by the target episodes, with true actions for the oracle and IDM pseudo-labels otherwise; ACT's 80/20 split is drawn over all of them. Pseudo-labels use the policy's own action space (absolute joint targets and gripper opening), and the exporter never reads the stored actions or states of a pseudo-labeled episode (R1, R3).

Table 3: Decisions that differ from the proposal.

| Topic | Decision | Proposal | Reason |
|---|---|---|---|
| Control rate | 10 Hz, at most 200 steps; chunk 20 (2 s) | 50 Hz, chunk 25 | One image per control step; the chunk keeps ACT's 2 s horizon |
| Actions | Absolute joint targets | Joint increments | ACT's convention; increments assume the state they were predicted from, so re-querying or averaging them fails |
| Execution | Temporal ensembling | Chunks open loop | Open loop gives 41% against 98% for the same policy (Table 13) |
| Initial state | Object yaw fixed at 0; $20 \times 20$ cm start area; fixed arm pose | OCBench defaults | As in ACT's simulated tasks; isolates the object shift |
| Expert | Deterministic: no detours, fixed speed, no $90^\circ$ grasp turns | OCBench's, adapted per object | 99% against 53% source success; turns are impossible for blocks (Section 2.2) |
| Goal | 12 cm square; inside, resting, released, still for 1 s | Within 4 cm | A visible goal and a strict success test |
| Policy | Original ACT code; checkpoint chosen by validation rollouts | Own ACT, fixed training steps | Our reimplementation underperformed; validation loss does not track success |
| Split | 80/20 train/validation by episode; test by rollouts on reserved seeds | 80/10/10 | Upstream ACT's split; success needs rollouts, not held-out demos |
| R6 | $< 100$ ms per call | $< 0.5$ s per chunk | Temporal ensembling queries the policy every step |

**Deviations from the proposal.** Table 3 lists the decisions that differ from the proposal. Most follow from one finding: our first policy, a reimplementation of ACT with the proposal's settings (joint increments, open-loop chunks, stochastic data), did not exceed 49% validation success on the source cube (Section 4). We replaced it with the original ACT and adopted its conventions.

### 2.4 Evaluation

The protocol was fixed in the repository's configuration before any result.

**Splits.** Source demonstrations are split by episode into 80% training and 20% validation (ACT's own random split). Fifty held-out source demonstrations (seeds 50,000+) are used only for the IDM's test error. Closed-loop evaluation uses two disjoint seed ranges shared by all methods and objects: validation rollouts (seeds 900,000+) and test rollouts (seeds 1,000,000+).

**Checkpoint selection.** Each ACT run saves the checkpoint with the lowest validation loss (*best*) and the final one (*last*). We run 100 validation rollouts on the *source* object for each and keep the one with higher success, with ties going to *best*. Selection never uses the target's actions or the target's success, so R1 holds for model selection too. The R2 decision (100 or 300 demonstrations) uses the same validation rollouts.

**Test.** Each selected checkpoint is evaluated once: 100 test rollouts per object, from the same initial states for every method. Each method is trained with three seeds; we report the mean and standard deviation across seeds and, for each success rate, the 95% Wilson score interval. For $k$ successes in $n$ rollouts, $\hat{p} = k/n$ and $z = 1.96$,

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}. \tag{6}$$

Intervals are necessary beyond sampling error: GPU rendering is not bit-exact, so rerunning the same seeds changes some outcomes (Section 4).

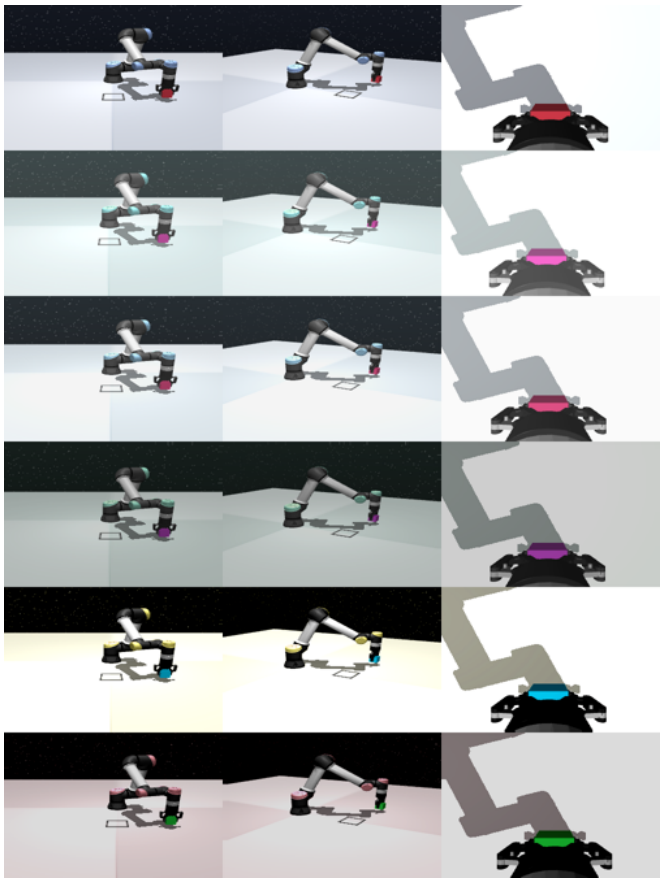


Figure 3: Augmentation of method (e). First row: an original training frame (front, side, wrist cameras). Other rows: five random draws.

**Metrics.** (i) Success rate on each target and on the source, from which R4 (Eq. 1) and R5 (Eq. 2) follow. (ii) Inference latency per policy call during rollouts, after warm-up, reported as the 95th percentile (R6). (iii) IDM error per action component on held-out source transitions and, with the true target actions, pseudo-label error on the target videos; the second is a diagnostic only. (iv) A classification of failed rollouts: never lifted, placed just outside the goal, lifted and ended far, inside but not released.

### 2.5 Alternative solutions

All alternatives are run on our own data with the same ACT architecture, training budget and evaluation (Table 4). Methods (a) and (d) anchor R4: the source-only floor and the oracle with true target actions. Comparing (b) and (c) isolates the effect of adapting the IDM, since they differ only in $\lambda$. Method (e) tests whether target videos are needed at all: it is the baseline with every training sample augmented [22]. The first setting, fixed before its results, jitters brightness, contrast and saturation by $\pm 30\%$, draws the hue over the full circle so that the red cube takes every colour, and shifts the image by up to 4 of 224 pixels; one random draw is applied to the three cameras alike (Figure 3). Validation and test images are not augmented. [TODO: Confirm the augmentation settings (team review pending).] Optional method (f) moves the alignment from the IDM to the policy's encoder [18].

Table 4: Methods compared. All use the same source demonstrations and ACT policy.

| | Method | Target data | Target labels |
|---|---|---|---|
| (a) | Source only | None | None |
| (b) | Unadapted IDM ($\lambda = 0$) | Videos | IDM pseudo-labels |
| (c) | DANN-adapted IDM (ours) | Videos | IDM pseudo-labels |
| (d) | Oracle | Demonstrations | True actions |
| (e) | Source only with augmentation | None | None |
| (f) | (b) with policy-encoder alignment (optional) | Videos | IDM pseudo-labels |

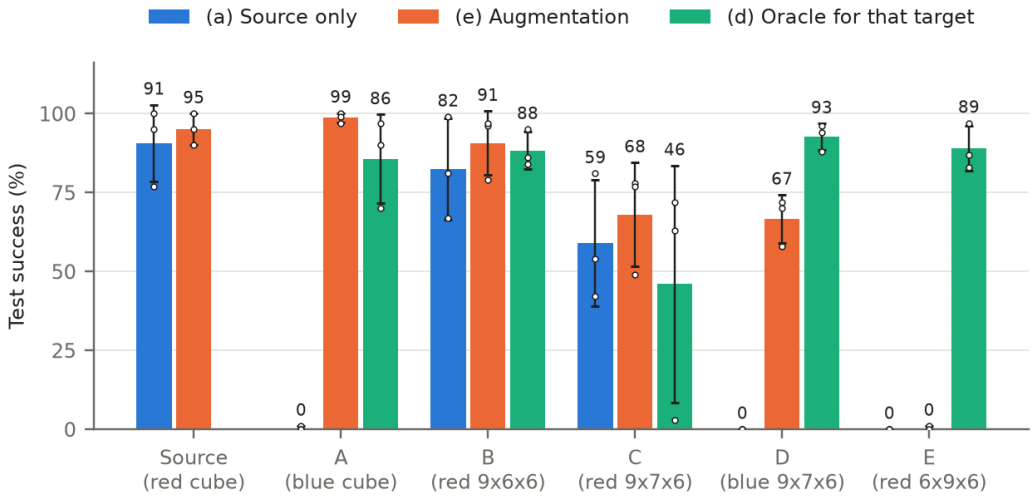



Figure 4: Test success per object: mean of three seeds (bars and numbers), standard deviation (lines) and each seed (dots), 100 rollouts per seed. The oracle is the one trained for that target. The low dot of the oracle for C is its seed 0 (Section 3, Oracle C). [TODO: Add (b) and (c).]

## 3 Results

Measured, with three seeds each: the source-only baseline (a) and the augmentation baseline (e) on every object, and the oracle (d) for every target. The methods that use target videos, (b) and (c), are pending. Figure 4 gives the overview; the tables give every seed.

### 3.1 Label budget (R2)

With 100 source demonstrations the selected checkpoint reaches 37% success in source validation rollouts, below the 80% of the rule. Every method therefore uses 300 source demonstrations.

### 3.2 Source-only baseline (a)

Table 5 reports the baseline on every object. It reaches $90.7 \pm 12.1\%$ on the source cube. It transfers to the red blocks (B 82.3%, C 59.0%) and fails on both blue objects (A 0.3%, D 0%), where it almost never lifts the object (1 of 199 failures did), and on E (0%), which it cannot grasp without turning. Inference takes 10 to 18 ms per call (95th percentile) over the 60 test runs of this report, well inside the R6 budget.

### 3.3 Oracle (d)

Table 6 reports the oracle. With the true actions of the target the policy reaches 85.7% on A, 92.7% on D and 89.0% on E, from 0.3%, 0% and 0%, and keeps the source within R5 in every case (87.0% to 96.3% against 90.7%). On B the oracle (88.3%) is within the baseline's spread ($82.3 \pm 16.0\%$), so B leaves little room for any method. On C the oracle's mean (46.0%) is below the baseline's (59.0%):

Table 5: Baseline (a): original ACT, images only, 300 source demonstrations. Test success over 100 rollouts per object and seed; 95% Wilson interval over the 300 pooled rollouts.

| Test object | Seed 0 | Seed 1 | Seed 2 | Mean | Std | Wilson 95% |
|---|---|---|---|---|---|---|
| Source (red cube) | 77% | 100% | 95% | **90.7%** | 12.1 | [86.8, 93.5] |
| A (blue cube) | 1% | 0% | 0% | **0.3%** | 0.6 | [0.1, 1.9] |
| B (red $9 \times 6 \times 6$ block) | 67% | 99% | 81% | **82.3%** | 16.0 | [77.6, 86.2] |
| C (red $9 \times 7 \times 6$ block) | 42% | 81% | 54% | **59.0%** | 20.0 | [53.4, 64.4] |
| D (blue $9 \times 7 \times 6$ block) | 0% | 0% | 0% | **0.0%** | 0.0 | [0.0, 1.3] |
| E (red $6 \times 9 \times 6$ block) | 0% | 0% | 0% | **0.0%** | 0.0 | [0.0, 1.3] |

Table 6: Oracle (d): source plus 200 target episodes with true actions. Test success over 100 rollouts per seed; 95% Wilson interval over the 300 pooled rollouts.

| Target | Tested on | Seed 0 | Seed 1 | Seed 2 | Mean | Std | Wilson 95% |
|---|---|---|---|---|---|---|---|
| A | A | 70% | 97% | 90% | **85.7%** | 14.0 | [81.2, 89.2] |
| A | Source | 77% | 97% | 87% | 87.0% | 10.0 | [82.7, 90.3] |
| B | B | 86% | 84% | 95% | **88.3%** | 5.9 | [84.2, 91.5] |
| B | Source | 94% | 91% | 82% | 89.0% | 6.2 | [85.0, 92.1] |
| C | C | 3% | 72% | 63% | **46.0%** | 37.5 | [40.4, 51.7] |
| C | Source | 67% | 100% | 94% | 87.0% | 17.6 | [82.7, 90.3] |
| D | D | 88% | 96% | 94% | **92.7%** | 4.2 | [89.1, 95.1] |
| D | Source | 96% | 100% | 93% | 96.3% | 3.5 | [93.6, 97.9] |
| E | E | 87% | 97% | 83% | **89.0%** | 7.2 | [85.0, 92.1] |
| E | Source | 94% | 91% | 94% | 93.0% | 1.7 | [89.5, 95.4] |

R4 is inconclusive for C by the rule of Section 2.4. On E the oracle is close to the expert's own success (93.9%): the turned grasp is learnable from 200 episodes, so E is a usable target.

**Oracle C, seed 0.** Seed 0 of the oracle for C reaches 3% on C, against 72% and 63% for seeds 1 and 2. It pulls the mean down to 46.0%; the median of the three seeds is 63%. We kept the run and examined it.

*What fails.* We measured where the gripper is, relative to the object's centre, when it first closes (10 validation rollouts per case, Table 7). The policy reaches the block at the right height but about 1 cm off-centre along the closing direction, more than the 0.75 cm that a 7 cm block allows: one finger lands on the block, the gripper closes empty and pushes the block. None of its 97 failed test rollouts lifted the block. Both of its checkpoints show the offset.

*It is not the test.* Run again on 100 validation rollouts of C, with other initial states, the three selected checkpoints give 1%, 75% and 61%.

*It is not the object.* Seeds 1 and 2 of the same configuration learn C. A policy trained on C alone (200 episodes, true actions, one seed; a diagnosis, not a method) reaches 87% on C in validation with its lowest-loss checkpoint. The oracle for D, the same block in blue, reaches 88% to 96% in its three seeds.

*The loss does not show it.* The validation L1 loss of seed 0 over the last 100 epochs is 0.058, against 0.059 for seeds 1 and 2, and its minimum is the lowest of the three. An offset of 1 cm at the grasp barely changes the mean action error but decides the rollout. The run was also weak on the source: 60% in validation against 97% and 98%.

The seed sets the weight initialisation and the sampling order only; we did not find which of the two produced the offset. The narrow clearance makes C fragile: the policy trained on C alone drops from 87% to 18% between its two checkpoints. The offsets of seeds 1 and 2 were not measured.

### 3.4 Augmentation (e)

Table 8 reports the source-only policy trained with augmented images. Without any target data it reaches 98.7% on the blue cube, above the oracle (85.7%), and 66.7% on the blue block D, from 0%. It does not cost source success (95.0% against 90.7%). On D it is level with the red block of the same

Table 7: Gripper position relative to the object's centre when the gripper first closes: lateral offset (mean $\pm$ std) and height, over 10 validation rollouts.

| Policy (seed) | Object | Lateral offset | Height | Success |
|---|---|---|---|---|
| Expert demonstrations | C | $0.0 \pm 0.1$ cm | 1.5 cm | – |
| Oracle C (0), selected | C | $+0.9 \pm 0.3$ cm | 1.5 cm | 0/10 |
| Oracle C (0), best | C | $+1.0 \pm 0.4$ cm | 1.5 cm | 0/10 |
| Oracle C (0), selected | Source | $+1.1 \pm 0.2$ cm | 1.5 cm | 6/10 |
| Oracle B (0) | B | $+0.6 \pm 0.3$ cm | 0.9 cm | 9/10 |
| Baseline (1) | C | $+0.1 \pm 0.4$ cm | 0.1 cm | 8/10 |
| Baseline (1) | Source | $-0.2 \pm 0.3$ cm | $-0.7$ cm | 10/10 |

Table 8: Augmentation (e): source only, augmented training images. Test success over 100 rollouts per object and seed; 95% Wilson interval over the 300 pooled rollouts.

| Test object | Seed 0 | Seed 1 | Seed 2 | Mean | Std | Wilson 95% |
|---|---|---|---|---|---|---|
| Source (red cube) | 95% | 90% | 100% | **95.0%** | 5.0 | [91.9, 96.9] |
| A (blue cube) | 97% | 100% | 99% | **98.7%** | 1.5 | [96.6, 99.5] |
| B (red $9 \times 6 \times 6$ block) | 96% | 79% | 97% | **90.7%** | 10.1 | [86.8, 93.5] |
| C (red $9 \times 7 \times 6$ block) | 78% | 49% | 77% | **68.0%** | 16.5 | [62.5, 73.0] |
| D (blue $9 \times 7 \times 6$ block) | 72% | 58% | 70% | **66.7%** | 7.6 | [61.2, 71.8] |
| E (red $6 \times 9 \times 6$ block) | 0% | 1% | 0% | **0.3%** | 0.6 | [0.1, 1.9] |

shape (C, 68.0%): once colour is handled, what remains is the 7 cm grasp, and there the oracle is 26 points higher (92.7%). On E it fails like the baseline (0.3%).

### 3.5 A target that changes the behaviour (E)

Because (e) solves the appearance shift, we looked for a target where it cannot help. We ran the trained checkpoints of (a) and (e) on four candidate objects (Table 9). The augmented policy handles every change of appearance, including a texture and gray levels that the jitter never produces. Only the block that needs another grasp defeats every policy; it became target E. E was chosen after seeing these results.

**Wrist angles and the IDM.** The IDM learns actions from source demonstrations only. In E's videos the last wrist joint spends 75.5% of the frames below $-1.2$ rad, where the 300 source demonstrations have 2.5% of theirs (Figure 5). We therefore recorded 100 more demonstrations of the *source cube* with every grasp turned by $90°$. Only the IDM trains on them; every policy keeps the 300 source demonstrations, so the results above are unchanged and all policies see the same labeled data. This stays within the problem: the data are labeled demonstrations of the source object, no target action is used (R1), and nothing in them says that E needs the turn; that comes from E's videos. The IDM thus sees 400 labeled source demonstrations and the policies 300. Two controls are open. [TODO: IDM label error on E with and without the turned demonstrations; source-only policy trained on the 400 demonstrations, tested on E.]

### 3.6 Checkpoint selection and training curves

Figure 6 shows, for every run, the source validation success of its two checkpoints. The lowest-loss checkpoint was selected in 8 of 21 runs and the last one in 13, with differences of up to 79 points. Figure 7 shows why the loss cannot choose: the validation loss flattens after about 1,000 epochs while success keeps changing. With augmentation the training loss stays close to the validation loss.

### 3.7 Inverse dynamics model

[TODO: Table: IDM error per action component on held-out source transitions, and pseudo-label error on each target, for $\lambda = 0$ (b) and DANN (c); three seeds. Figure: IDM training curves.]

Table 9: Screening of candidate targets: success of the three seeds of (a) and (e) over 20 validation rollouts each (about $\pm 20$ points per value). No training and no test seeds were used.

| Candidate | (a) Source only | (e) Augmentation |
|---|---|---|
| Cube with a black and white checker texture | 0, 0, 0% | 100, 100, 85% |
| Black cube | 0, 0, 0% | 80, 70, 75% |
| White cube | 0, 0, 0% | 80, 90, 60% |
| Red $6 \times 9 \times 6$ block (9 cm side across the gripper) | 0, 0, 0% | 0, 0, 0% |

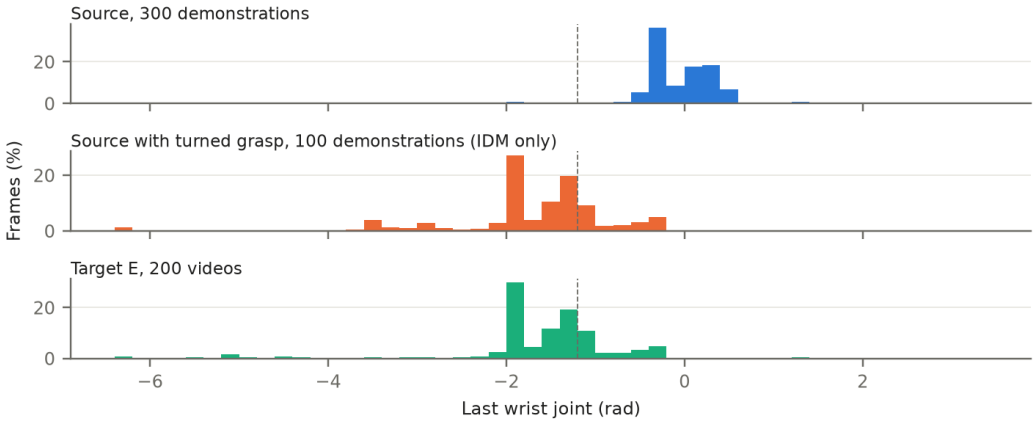



Figure 5: Angle of the last wrist joint over all frames of three collections. The dashed line marks $-1.2$ rad.

### 3.8 Comparison with alternatives

Table 10 holds the comparison on the C4 constraints (C9, C10), and Table 11 the requirement R4 for (e).

The augmentation baseline meets R4 on A, B and D without target data, and R5 on the source. On A it leaves nothing to gain. The room for methods that use target videos is on D, 26 points between (e) and the oracle, and on E, 89 points, where (e) gains nothing. [TODO: R4 ($G_T$) and R5 per target for (b) and (c); (b) against (c); (c) against (e); latency of every method (R6); negative transfer if alignment hurts.]

### 3.9 Results on earlier data

The runs below used the data before the grasp-turn change (Section 2.2), with one seed. They motivated the choice of data and of the execution scheme and are not comparable with the tables above.

**Execution ablation.** Table 13 compares four ways of executing the same policy's 20-action chunks. Executing each chunk open loop, as the proposal planned, drops success from 98% to 41%, and re-querying more often without averaging is worse.

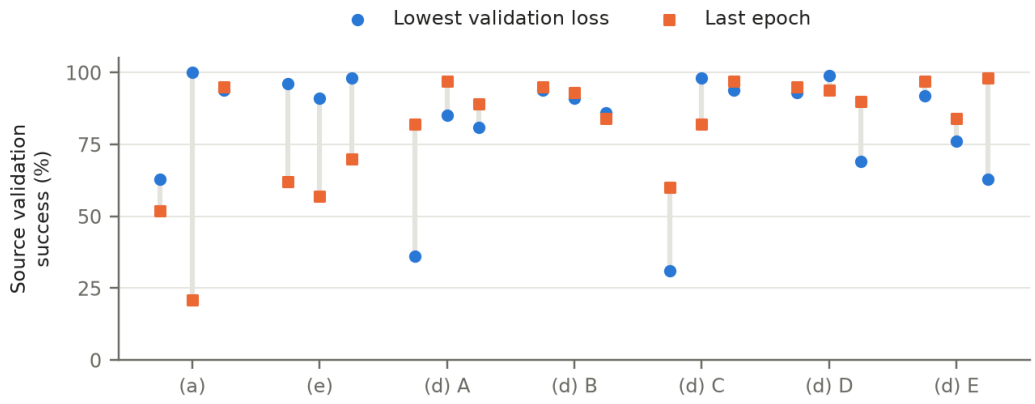



Figure 6: Checkpoint selection: success in 100 source validation rollouts of the lowest-validation-loss checkpoint and of the last one, three seeds per method. The higher one is tested.

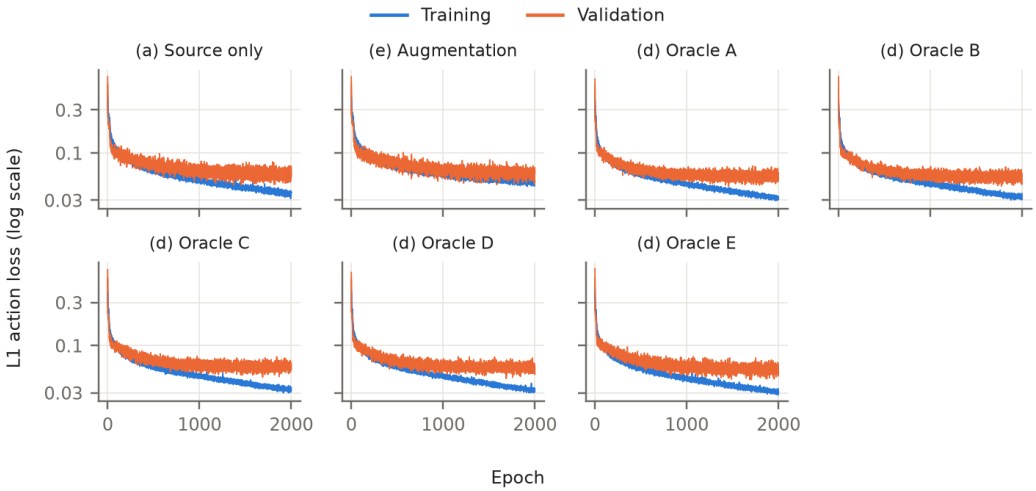



Figure 7: ACT's L1 action loss per epoch, three seeds per panel. [TODO: Add the policies of (b) and (c).]

Table 10: Test success on each target and on the source (mean over three seeds, 100 rollouts each). (a) and (e) use no target, so their source column is one number; for (d) it is the range over the oracles for A to E.

| Method | A | B | C | D | E | Source |
|---|---|---|---|---|---|---|
| (a) Source only | 0.3% | 82.3% | 59.0% | 0.0% | 0.0% | 90.7% |
| (e) Augmentation | **98.7%** | **90.7%** | **68.0%** | 66.7% | 0.3% | 95.0% |
| (b) Unadapted IDM | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| (c) DANN-adapted IDM | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] | [TODO: ] |
| (d) Oracle | 85.7% | 88.3% | 46.0% | **92.7%** | **89.0%** | 87.0–96.3% |

Table 11: R4 for the augmentation baseline (e): fraction of the oracle's gain over the source-only policy, from the means of Table 10.

| Target | Oracle gain | (e) gain | $G_T$ | R4 ($\geq 0.5$) |
|---|---|---|---|---|
| A | 85.4 points | 98.4 points | 1.15 | Met |
| B | 6.0 points | 8.4 points | 1.40 | Met; the gap is within the seed spread |
| C | −13.0 points | 9.0 points | – | Inconclusive (oracle below baseline) |
| D | 92.7 points | 66.7 points | 0.72 | Met |
| E | 89.0 points | 0.3 points | 0.00 | Not met |

Table 12: Earlier data, one seed: original ACT, 300 source demonstrations. Validation and test success over 100 rollouts each; 95% Wilson intervals in brackets. The robot-state run breaks R3 and is a reference only.

| | | Validation (source) | | Test | |
|---|---|---|---|---|---|
| Data | Input | best | last | Source | Target A |
| Stochastic | Images | 40% | 53% | 53% [43, 62] | 0% [0, 4] |
| Deterministic | Images | 61% | 98% | 99% [95, 100] | 7% [3, 14] |
| Deterministic | Images + state | 44% | 66% | 69% [59, 77] | 1% [0, 5] |

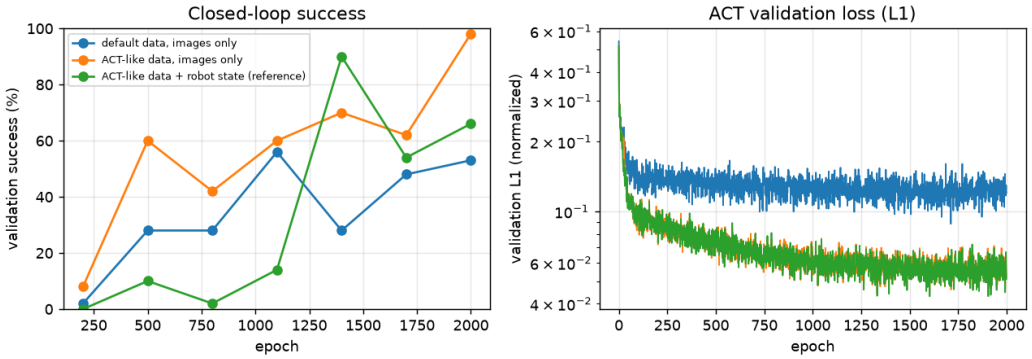



Figure 8: Earlier data: validation success per epoch (50 rollouts; 100 at the last epoch) and ACT's validation loss for the three runs of Table 12. The loss is smooth while success varies between nearby checkpoints.

Table 13: Execution of the 20-action chunks: earlier deterministic-data policy, 100 source validation rollouts, one seed.

| Execution | Success |
|---|---|
| Temporal ensembling, $k = 0.01$ (used everywhere) | **98%** |
| Open loop: whole chunk, then re-query (proposal) | 41% |
| Re-query every 10 steps | 32% |
| Re-query every step, first action only | 1% |

## 4 Discussion and Conclusions

**Were the constraints met?** R2 is decided (300 demonstrations). R3 and R6 are met by every policy measured: images only, at most 18 ms per call. The augmentation baseline meets R4 on A, B and D and R5 on the source. [TODO: R1 and R3 for the IDM pipeline; R4 and R5 per target for (b) and (c).]

**Appearance does not need target videos.** The source-only policy fails on everything that is not plain red, and colour jitter on the same source data removes that failure, on the blue cube and on objects it was never tuned for (texture, black, white). On an appearance shift, target videos compete with an alternative that needs no target data and already exceeds the oracle on A.

**Behaviour does.** Augmentation changes what the policy tolerates in the image, not what it does. When the target needs another grasp it gains little (D: 66.7% against an oracle of 92.7%) or nothing (E: 0.3%). These are the targets where action-free videos can add something, and where (b) and (c) should be judged. [TODO: Interpret with the results of (b) and (c) on A, D and E.]

**Demonstration variability.** On the earlier data the same code and budget gave 99% or 53% depending on how the demonstrations were generated. The ACT paper reports the same direction between scripted and human demonstrations [3].

**Checkpoint and seed variance.** Neither the lowest validation loss nor the last epoch is a reliable choice: in the 18 runs of Figure 6 the selected checkpoint was *best* 8 times and *last* 10 times, with differences of up to 79 points. Selection needs closed-loop rollouts. Across seeds, source success ranges from 77% to 100% and one oracle run failed (C, seed 0), so three seeds give wide intervals; the Wilson intervals pool the rollouts of the three seeds and do not include that spread. [TODO: Cite prior reports of this effect (not yet in the bibliography).]

**What went wrong.** We first wrote our own image-only ACT with the proposal's settings: joint increments, chunks of 10 executed open loop, stochastic demonstrations. With 100, 200 and 300 demonstrations it reached 23%, 49% and 31% validation success (one seed each); even from training initial states it succeeded only 60–70% of the time, and most failures were missed grasps. Temporal ensembling made it worse (21% and 14% for 200 and 300 demonstrations), because averaging increments predicted from different states is not well defined. We replaced it with the original ACT, absolute targets and deterministic demonstrations. One oracle run for target C also failed (Section 3), and our first choice of targets could not answer the research question, because augmentation solves them (Section 3.5). [TODO: Other problems met with the IDM and DANN.]

**Limitations.**

- **Simplified task.** The object's yaw, the arm's start pose and the expert's speed and grasp angle are fixed, and the start area is $20 \times 20$ cm. The stochastic reference (53%) shows that the baseline does not handle stock OCBench variability.
- **Simulation only.** Target videos come from the same simulator and expert as the source, so the only shift is the object.
- **Protocol change.** The expert lost its grasp turns after the 100-demonstration result; R2 was not revisited.
- **Oracle.** Every checkpoint, the oracle's too, is selected on source validation rollouts, which may understate the oracle on its target. The 0.75 cm clearance makes C fragile, and its oracle gives no usable gap.
- **Target E.** It was designed after the augmentation results. Its expert succeeds in 93.9% of attempts, which bounds the oracle. The IDM trains on 100 source demonstrations that the policies do not see.
- **Augmentation.** One setting, not tuned; its hue range covers every colour, the favourable case. The screening of candidates used 20 rollouts per value.
- **Evaluation noise.** GPU rendering is not bit-exact: two renders of the same state can differ by one intensity level in a few pixels. With our earlier reimplementation this flipped 11 of 40 episode outcomes when the same seeds were rerun.

- **Adversarial alignment.** DANN aligns marginal feature distributions; it does not guarantee that actions remain readable from the aligned features [19], and it is weaker for regression than for classification [20]. [TODO: Outcome from the per-component IDM errors.]
- **Operations.** The original ACT code cannot resume, so a preempted cloud run restarts from zero.

**Advantages.** [TODO: Advantages supported by the final results (no target labels, no simulator state, IDM offline so the deployed policy costs the same as the baseline).]

**Conclusion.** [TODO: Answer the research question: do action-free videos, labeled by an adapted IDM, close the gap to the oracle on each target while retaining the source?]

## Contributions

All members collaborated on and reviewed the report. [TODO: Update to what each member did.]

| Member | Role | Tasks |
|---|---|---|
| Brik Meza | Data and evaluation | Simulation, expert demos, datasets, splits, rollouts, metrics |
| Myriam Dardour | Adversarial adaptation | Discriminator, adversarial training, tuning, experiments |
| Zheng Ye Zhang | Inverse dynamics model | IDM architecture and training, pseudo-labeling, error analysis |
| Gonzalo Millán García | Baselines and alternatives | Policy training, baselines, alternative methods, comparisons |

## Generative AI disclosure

We used ChatGPT and Codex (OpenAI) and Claude and Claude Code (Anthropic), following McGill's guidelines on generative AI in research. They assisted with (i) the literature search and bibliography formatting; (ii) document structure and draft prose of the proposal and this report; (iii) software: simulator changes in our OCBench fork, seeded data collection and replay checks, dataset publishing, the training and evaluation infrastructure, a first ACT reimplementation and its diagnostics, the adapter that runs the original ACT, and tests; (iv) running and summarizing the baseline, oracle and augmentation experiments that a team member approved, including the screening of candidate targets and the definition of target E; and (v) the figures of this report. We used them to cover more engineering than a four-person course project otherwise allows, so that the team's time went to design decisions and analysis. The team directed the work, made the design decisions (scene, data sizes, protocol, ACT settings) and approved every paid run. Verification: automated tests, exact replay of stored episodes, end-to-end training and evaluation tests, and an independent review of the ACT adapter against the upstream code; every reported number comes from logged runs. The team revised the text and checked claims and citations against the sources. [TODO: Update from docs/ai-use-log.md before submission, including the IDM and DANN work.]

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